

Clustering Algorithms Comparison

Introduction

Clustering is an Unsupervised Machine Learning technique used to group similar data points together without relying on labeled data. The goal is to discover hidden patterns, structures, or relationships within a dataset.

Unlike supervised learning, clustering does not require predefined target labels. It is commonly used in customer segmentation, recommendation systems, image analysis, anomaly detection, and market research.

This project explores and compares three popular clustering algorithms:

- K-Means Clustering
 - Hierarchical Clustering
 - DBSCAN
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1. K-Means Clustering

Overview

K-Means is a centroid-based clustering algorithm that divides data into a predefined number of clusters (K).

Working

1. Select the number of clusters (K).
2. Initialize cluster centroids.
3. Assign each data point to the nearest centroid.
4. Recalculate centroids.
5. Repeat until cluster assignments stabilize.

Advantages

- Simple and easy to implement.
- Fast and computationally efficient.
- Works well with well-separated clusters.

Limitations

- Requires the number of clusters in advance.
 - Sensitive to outliers.
 - Performs poorly with irregular cluster shapes.
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2. Hierarchical Clustering

Overview

Hierarchical Clustering builds a hierarchy of clusters by successively merging or splitting data points.

In this project, Agglomerative Clustering (bottom-up approach) is used.

Working

1. Treat each data point as an individual cluster.
2. Merge the closest clusters.
3. Continue merging until the desired number of clusters is reached.

Advantages

- No random initialization required.
- Helps visualize relationships among clusters.
- Suitable for smaller datasets.

Limitations

- Computationally expensive for large datasets.
 - Less scalable compared to K-Means.
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3. DBSCAN

Overview

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) groups data points based on density.

Key Parameters

- **eps:** Radius around a data point.
- **min_samples:** Minimum number of neighboring points required to form a cluster.

Working

1. Identify dense regions in the dataset.
2. Expand clusters from dense regions.
3. Label isolated points as noise.

Advantages

- Detects clusters of arbitrary shapes.
- Handles noise and outliers effectively.
- Does not require specifying the number of clusters.

Limitations

- Sensitive to parameter selection.
 - May struggle with datasets having varying densities.
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Comparison of Clustering Algorithms

Feature	K-Means	Hierarchical	DBSCAN
Clustering Type	Centroid-Based	Hierarchy-Based	Density-Based
Requires Number of Clusters	Yes	Usually Yes	No
Handles Noise	No	Limited	Yes
Computational Speed	Fast	Moderate	Moderate
Scalability	High	Medium	Medium
Arbitrary Cluster Shapes	No	Partial	Yes
Best For	Well-Separated Data	Understanding Relationships	Complex Shapes & Noisy Data

Conclusion

K-Means, Hierarchical Clustering, and DBSCAN are widely used clustering algorithms, each with unique strengths and limitations. K-Means is fast and efficient for well-defined clusters, Hierarchical Clustering is useful for understanding cluster relationships, and DBSCAN excels at detecting complex-shaped clusters and handling noise. The choice of algorithm depends on the dataset characteristics and the specific problem being addressed.